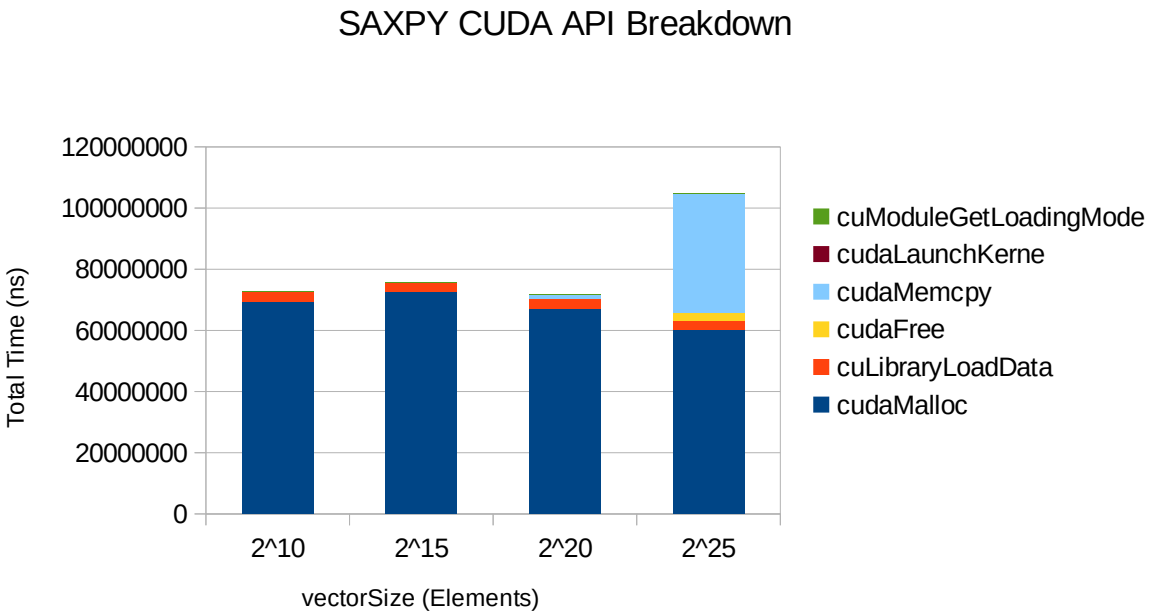
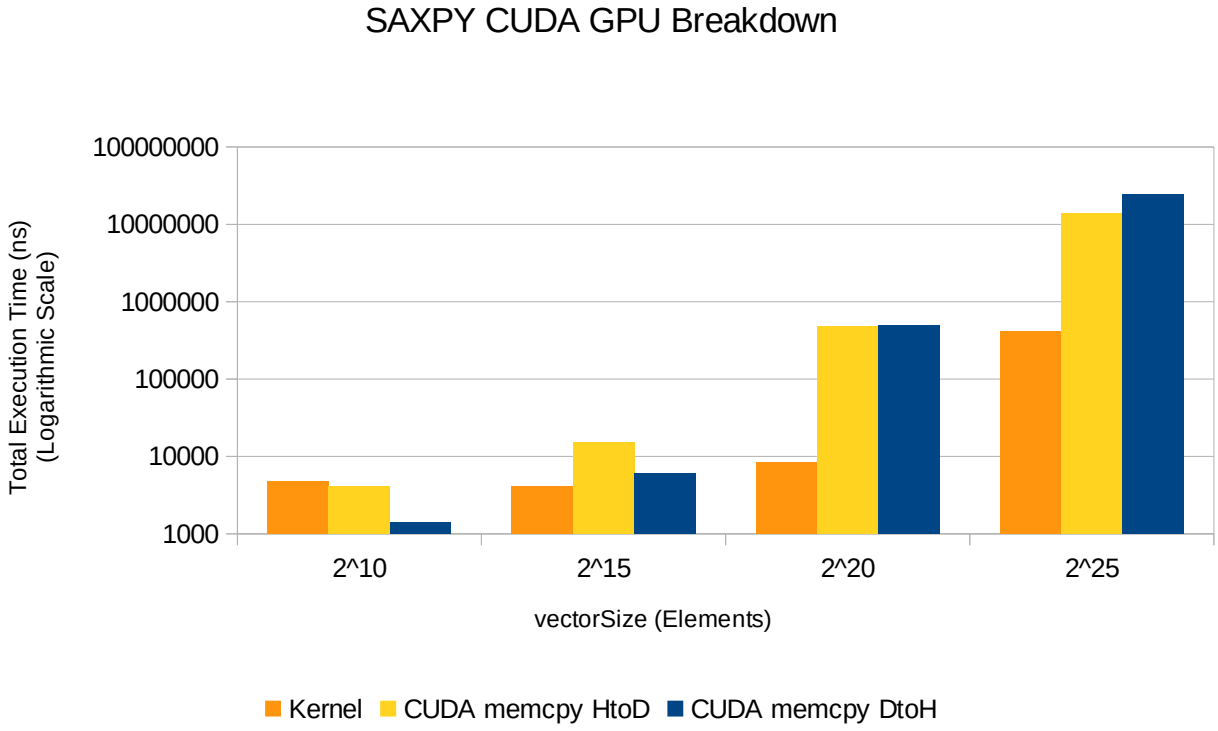
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SAXPY Profiling Results:



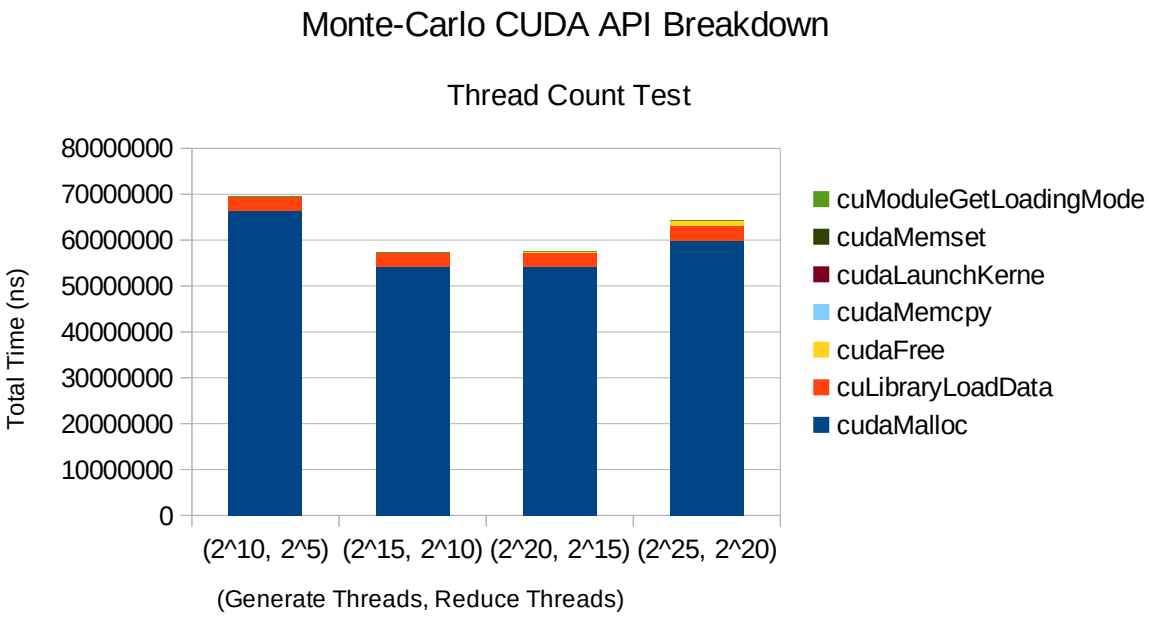


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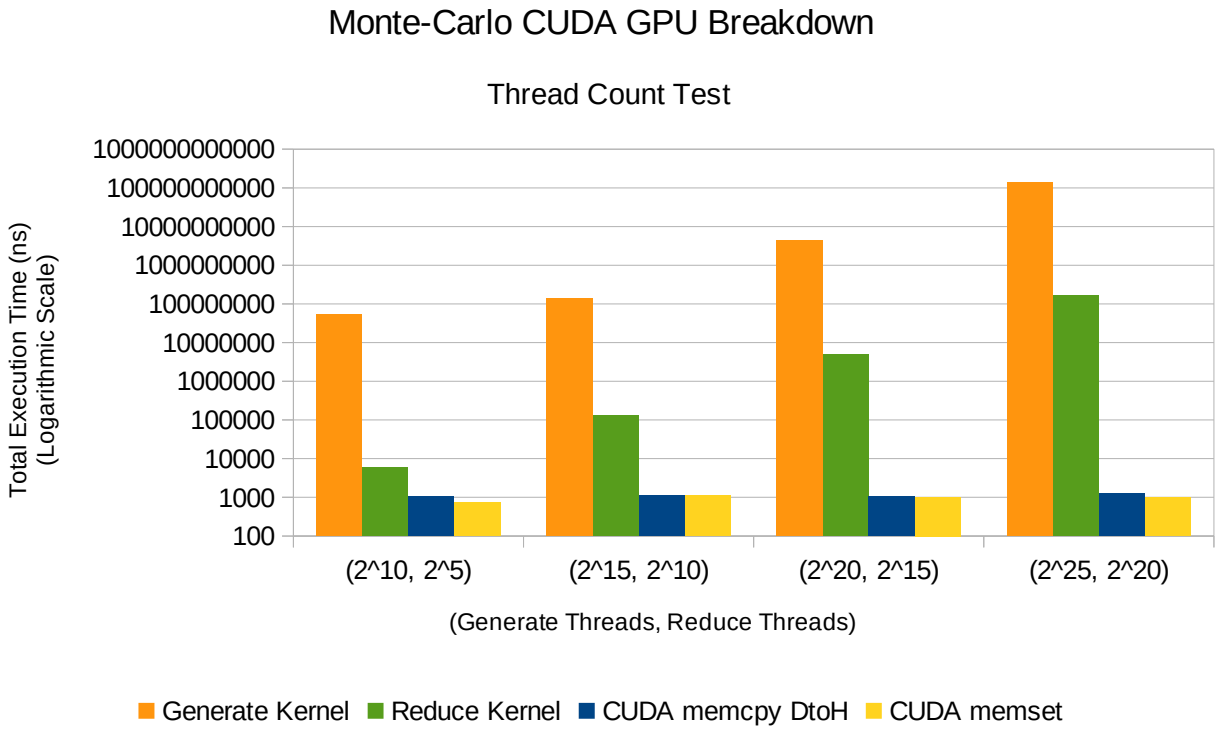
Observations:

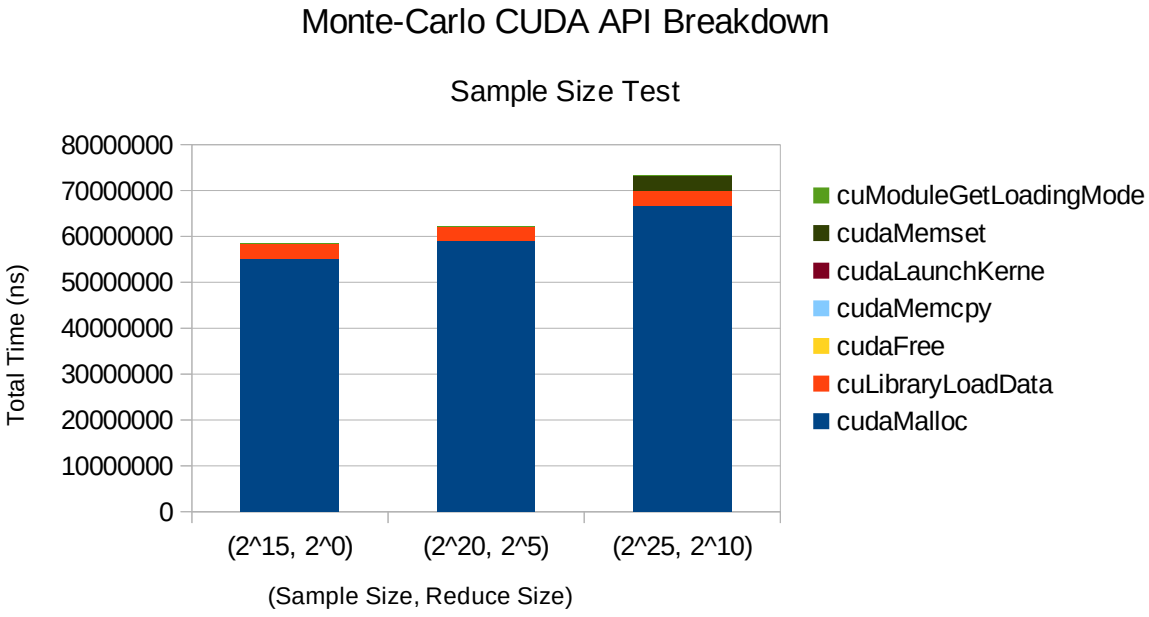
The SAXPY kernel was profiled with four different sizes of vectors ranging from 2^{10} to 2^{25} entries. Based off the test data, it appears that a majority of the execution time is spent on roughly constant-time API calls such as cudaMalloc. Only with 2^{25} entries does the execution time of the kernel and associated memory transfers become appreciable, though it is still less than the API calls. That said, based on the rate of growth, it can be assumed that the execution time of the kernel and associated memory transfers will exceed the API calls given a large enough dataset. Another observation is that the total execution time of memory transfers generally grows faster than the kernel's, so this program appears to be memory-bound in that sense. Note that cudaDeviceSynchronize was omitted from the API breakdown as it was used to isolate the kernel's execution time from other statistics (for clarity). The same will apply to the Monte-Carlo results.

Monte-Carlo Profiling Results:

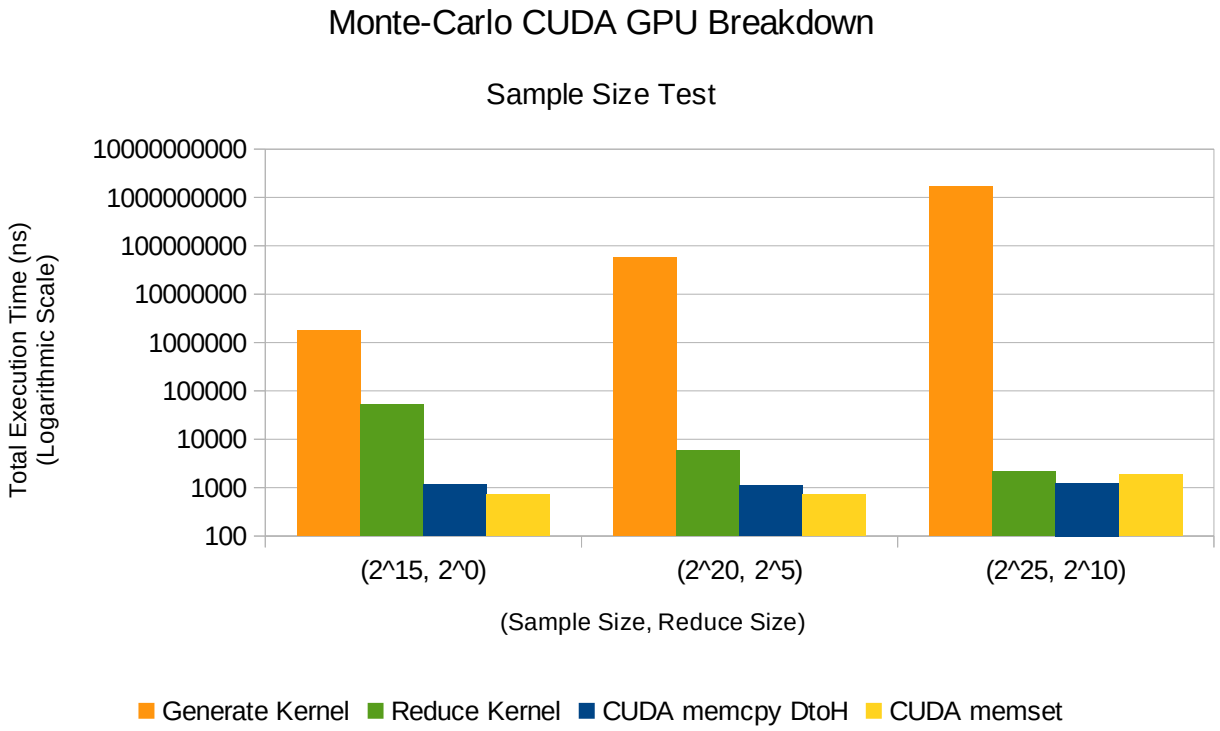


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Observations:

For the Monte-Carlo algorithm, the total number of generate threads and reduce threads were tested and increased together, while the sample size and reduce size were increased together in a separate test. This was done for clarity, as well as the fact that the parameters in each group are somewhat linked to each other. Note that the values/parameters not tested in each graph are left at default values. Once again, it appears that the execution time of the API calls are relatively constant, with some calls like cudaMalloc growing linearly. However, unlike SAXPY, the time spent in kernels is much more comparable to the execution time of API calls for lower test parameters. Additionally, the kernel execution time appears to grow much faster (especially generatePoints) than the memory transfers or API calls. Thus, it can be said that this algorithm is more compute bound than SAXPY is.